



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER I) – June, 2018
INTRODUCTION TO STATISTICS - ST 101

Answer **all** questions in the space provided.

Time Allowed: **TWO Hours.**

1. a) What is the difference between a **parameter** and a **statistic**?

- b) Give an example of when a **statistic** might preferred by a researcher over a parameter.

- c) List five methods of collecting data.

- d) Classify each of the following as nominal, ordinal, interval or ratio data.

- | | |
|---|----------------------|
| i. Brand of chocolate you prefer | <input type="text"/> |
| ii. Year of your birth (eg. 1994, 1995) | <input type="text"/> |
| iii. Weight of your pet | <input type="text"/> |
| iv. Degrees earned by your teachers | <input type="text"/> |
| v. Egg sizes (Small, Medium, Large, Extra Large, Jumbo) | <input type="text"/> |

2. a) Under what conditions might a median be a better measure of the center of data set than the mean?

- b) Typically we consider quantitative data that is symmetric about the mean. If we have a data set that has a few extreme high values, then

- i. How is the data skewed?

- ii. Would you prefer to use a mean or median? Justify your answer.

- c) The following sample data of the number of communications are taken from logs of communication with Distance Education students.

5, 9, 5, 23, 27, 55, 34, 7, 30, 15, 22, 60, 14, 52, 297, 8, 51, 15, 51, 35, 15, 39, 137, 43, 38, 14, 93, 7

- i. Compute the mean.

- ii. Find the standard deviation.

- iii. Determine the five number summary: Min, Max, Q_1 , Q_2 and Q_3 .

- iv. Is mean or median, a better representation of the central tendency? Explain.

3. a) What is the fundamental difference between a bar chart and a histogram?

- b) Create a stem-and-leaf plot for the following data set and interpret it.

45, 54, 66, 92, 85, 77, 43, 64, 72, 88, 95, 81, 100

- c) The distribution by age in Sri Lanka in 2014 is given in the table below.

Age	Percentage of population	Age	Percentage of population
0-5	7	35-45	15
5-15	14	45-55	14
15-20	7	55-65	10
20-25	7	65-75	6
25-30	7	75 and over	6
30-35	7		

- i. Draw the corresponding histogram. Notice that the class intervals include the left endpoint, not the right one. The interval for “75 and over” can be ended at 85.

- ii. Are there more 23-year-olds, or more age 63-year-olds?

- iii. What is the percentage of people age 35 or more ?

4. a) State the Empirical Rule.

- b) The weights of adorable, fluffy kittens are normally distributed with a mean of 3.6 pounds and a standard deviation of 0.4 pounds. Answer the following questions.

- i. What percent of adorable, fluffy kittens weigh between 2.8 and 4.8 pounds?

- ii. What percent of adorable, fluffy kittens weigh less than 2.4 pounds?

- iii. What value corresponds to a 97.5th percentile of kitten weights?

4. c) Suppose a distribution has mean $\mu = 17.4$ and standard deviation $\sigma = 3.2$. What does Chebyshev's Theorem tell us about the probability a randomly selected value is between 12.6 and 22.2?

- d) Suppose a distribution has mean $\mu = 111$ and standard deviation $\sigma = 7.6$. If Chebyshev's Theorem tells us that 81.1% of the values are between a and b (symmetrical about the mean), then what are these limits?

5. Following table shows the life expectancy for an individual born in Sri Lanka in certain years.

Year of Birth	Life Expectancy (in Years)	XY	X^2	Y^2
1930	59.7			
1940	62.9			
1950	70.2			
1965	69.7			
1973	71.4			
1982	74.5			
1987	75			
1992	75.7			
2010	78.7			

- a) Identify the independent variable and the dependent variable.

Independent variable:
Dependent variable:

- b) Draw a scatter plot of the ordered pairs.

- c) Calculate the least squares line by completing the above table and write the equation in the form of: $\hat{y} = \beta_0 + \beta_1 x$.

- d) Find the estimated life expectancy for an individual born in 1950 and for one born in 1982.

- e) Why aren't the answers to part (d) the same as the values in Table that correspond to those years?

- f) Use the two points in part (d) to plot the least squares line on your graph from part (b).

- g) Find the correlation coefficient. Is there a linear relationship between the year of birth and life expectancy?

- h) Using the least squares line, find the estimated life expectancy for an individual born in 1850. Does the least squares line give an accurate estimate for that year? Explain why or why not.